HardwareMatch
Project Document

Web application for exploring technology products and generating hardware configurations

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1. General description

HardwareMatch is a web application that lets users browse a catalog of technology components and products, explore alternatives through filters and sorting criteria, and get hardware configuration proposals based on a budget and intended use.

The system also maintains a directory of stores and external links associated with the products. This lets users visit suppliers' pages to find commercial information or purchase directly on their platforms.

The project does not aim to become an online store or to process payments. Its main purpose is to make it easier to explore technology products and to support decision-making through recommendations based on rules defined by the development team.

2. Problem statement

Selecting technology components and products can be complex for people who lack the technical knowledge needed to evaluate specifications, compatibility, and available alternatives. In addition, comparing products means consulting multiple catalogs and websites, interpreting technical characteristics, and determining whether the options found fit a budget and a specific need.

When a person wants to build a computer or choose technology products for a specific purpose, they may have difficulty:

- Identifying the right components for their needs.
- Understanding and comparing technical specifications.
- Finding technically compatible combinations of components.
- Distributing a budget among the different components of a computer.
- Consulting different stores or suppliers to find a product.
- Organizing the alternatives and keeping a configuration proposal to review later.

Suppliers' commercial information can change over time. Visiting external links does not guarantee that a product is still available or that it keeps a given price.

2.1. Research question

How can the exploration of technology products and the selection of hardware configurations that meet a user's needs and budget be made easier, taking into account technical criteria, compatibility between components, and links to different suppliers?

3. Justification

The project proposes to centralize descriptive and technical information about technology products, to offer search and filtering tools, and to build a recommendation mechanism that generates justified alternatives according to the user's needs.

The scope is limited to a functional product for academic purposes. It is not intended to guarantee commercial availability, offer real-time market prices, process transactions, or operate as a commercial service in production.

4. Proposed solution

Develop a web application that integrates the following capabilities:

1. **User management and access:** allow users to register and authenticate, with different permissions depending on their role.

2. **Catalog administration:** allow the administrator to manage products, categories, brands, specifications, and publication status.

3. **Product exploration:** allow searches, combinable filters, and sorting by criteria such as name, category, brand, reference price, and registration date.

4. **Supplier directory:** register stores and multiple external links per product, including each store's name and logo.

5. **Hardware configurator:** collect the budget and intended use, evaluate eligible products, and generate at least two alternatives when enough data is available.

6. **Recommendation report:** present the selected components, their registered characteristics, the explanation of the proposal, and the available external links.

7. **Configuration persistence:** keep the requests and results so that users can consult them again.

The first version will use explicit business rules and a deterministic selection and scoring algorithm. Generative artificial intelligence and integration with the stores' commercial APIs are not required to complete the initial scope.

4.1. Price handling

The system may use internal reference prices, managed manually, to estimate the cost of a configuration. These values do not necessarily represent the suppliers' current prices.

In the first version, the system will not query or store external prices in real time. The external links will let users consult the supplier's information directly. Recommendations must warn that prices, availability, and commercial conditions must be verified on the destination site.

5. Objectives

5.1. General objective

Develop a web application to explore technology products and generate hardware configuration proposals according to the user's needs and budget, by means of selection rules, compatibility validations, and links to external suppliers.

5.2. Specific objectives

- Analyze the problem, identify the actors, and establish the functional and non-functional requirements.
- Model the system's domain and main interactions.
- Design a normalized relational database that represents users, roles, products, categories, brands, stores, links, and recommendations.
- Implement a REST API organized by responsibilities, with input validation, consistent error handling, and access controls.
- Build a web interface to manage and browse the catalog.
- Implement filters, sorting, and pagination in catalog queries.
- Develop a rule-based algorithm to generate hardware alternatives, considering the available data and technical constraints.

- Allow users to consult the external links associated with each product.
- Verify the main flows through tests and technical documentation.

6. Project scope

6.1. In scope

- Web application with administrator and customer roles.
- Registration, authentication, and basic user profile management.
- Administrative management of products, categories, brands, and stores.
- Recording of the products' technical specifications.
- Association of multiple external links with a single product.
- Catalog with search, combinable filters, sorting, and pagination.
- Form to define the budget and intended use.
- Compatibility rules for the combinations covered by the catalog.
- Generation of two configuration alternatives when the data allows it.
- Configuration reports and persistence of requests and results.
- External links to the pages of stores, distributors, or manufacturers.
- Tests, API documentation, and a reproducible development environment setup.

6.2. Out of scope

- Buying, selling, or paying for products within the platform.
- Management of inventory, shipping, returns, or supplier warranties.
- Integration with the Amazon, Mercado Libre, or other merchants' APIs in the first version.
- Automatic retrieval of prices, stock, promotions, or delivery times.
- Any guarantee that all linked products are available or have the registered reference price.

- Generative artificial intelligence or machine learning models to produce recommendations.
- Native mobile application.
- Deployment as a commercial service, production operation, or availability commitments.
- Compatibility recommendations for every product on the market; validation is limited to the rules and specifications implemented.

7. System actors

7.1. Administrator

Responsible for maintaining the system's data and controlling the catalog.

Responsibilities and capabilities:

- Log in to the system.
- Create, view, update, and deactivate products.
- Manage categories and brands.
- Record and update technical specifications.
- Register stores, logos, and external links.
- Review and correct catalog information.
- Maintain the internal reference prices, if this functionality is included in the implementation.
- Consult the available administrative information.

7.2. Customer or registered user

The person who uses the browsing and recommendation features.

Responsibilities and capabilities:

- Register and log in.
- View and update the basic data of their profile.
- Browse the product catalog.
- Search, filter, and sort results.
- View the technical details of a product.
- Define a budget and an intended use.
- Request configuration recommendations.
- Compare the generated alternatives.
- View the components included in each proposal.
- Open the suppliers' external links.
- View saved requests and reports, according to the access rules.

7.3. Unauthenticated visitor

A person who accesses the application without having logged in.

In the initial scope, visitors will be able to view the public presentation pages and, if so defined during the analysis, browse the public catalog. To request and save recommendations, they must authenticate.

Public access to the catalog is a decision that will be confirmed when the requirements are detailed; it must not be taken as authorization to access administrative operations.

7.4. External systems: stores and manufacturers

Amazon, Mercado Libre, technology stores, and manufacturers' official pages are external systems, not internal users of the application.

The system will store links registered by the administrator and will let users navigate to them. In the first version, these systems will not automatically send data to HardwareMatch, and no API integration with them will be implemented.

8. Functional modules

8.1. Authentication and users

Manages registration, login, authentication, profile, and access permissions.

8.2. Catalog administration

Manages products, categories, brands, technical specifications, and publication status.

8.3. Store and external link management

Manages the registered suppliers and the relationship between each product and its external links. A product may have several links, and a store may be related to multiple products.

8.4. Catalog search and browsing

Implements text search, combinable filters, sorting criteria, and pagination. Queries must be validated in the backend and must not depend exclusively on the interface.

8.5. Configurator and recommendation engine

Collects the user's preferences, selects candidates from the catalog, evaluates the compatibility rules, and calculates scores to build alternatives.

The rules must be explicit and verifiable. When required specifications are missing, or there are not enough compatible components, the system must communicate the

limitation instead of generating a configuration presented as valid.

8.6. Reports and saved configurations

Stores the user's request and the generated alternatives, together with their associated components. It allows the results to be consulted later and keeps the relationship between a request and its report.

8.7. Infrastructure and quality

Covers the setup of the development environment, database migrations, environment variables, tests, API documentation, and instructions for running the project.

9. Initial functional requirements

These requirements are preliminary and will be refined through use cases and acceptance criteria.

Code	Requirement
FR-01	The system will allow users to register and will authenticate their credentials.
FR-02	The system will distinguish between administrator and customer permissions.
FR-03	The administrator will be able to manage products, categories, brands, and specifications.
FR-04	The administrator will be able to manage stores and their presentation data.
FR-05	The administrator will be able to associate multiple external links with a product.
FR-06	The user will be able to browse the catalog and view a product's details.
FR-07	The user will be able to combine filters and sort the catalog results.
FR-08	The system will paginate the results of catalog queries.

Code	Requirement
FR-09	The user will be able to define a budget and an intended use for a recommendation request.
FR-10	The system will evaluate candidates according to defined technical rules and budget constraints.
FR-11	The system will generate at least two alternatives when enough data exists to do so validly.
FR-12	The system will explain the selected components and show their available external links.
FR-13	The user will be able to view the saved requests and configurations they have access to.
FR-14	The system will prevent users from accessing or modifying resources they are not authorized to use.
FR-15	The system will distinguish internal reference prices from external commercial prices.

10. Initial non-functional requirements

- **Maintainability:** organize the code into modules with clear responsibilities and low coupling.
- **Security:** store passwords using appropriate hashing algorithms, protect routes, and validate authorization in the backend.
- **Integrity:** define primary keys, foreign keys, constraints, and consistency rules in the database.
- **Validation:** validate input data in both the interface and the backend where applicable.
- **Error consistency:** use appropriate HTTP responses and a uniform structure to communicate errors.
- **Usability:** present forms, filters, specifications, and reports in an understandable way.

- **Testing:** verify business rules, permissions, filters, and relevant configurator scenarios.
- **Documentation:** describe the installation, configuration, endpoints, and main design decisions.
- **Reproducibility:** allow the team to run the project in a documented development environment.
- **Traceability:** relate requirements, use cases, design, implementation, and tests.

No quantitative performance or availability targets typical of a production system are set, since they are not part of the course scope.

11. Risks and pending decisions

Topic	Risk or pending decision
Catalog quality	The recommendations will only be as reliable as the registered specifications.
Compatibility	It must be defined which compatibilities will be validated and what information each one requires.
Budget	It must be decided how to record, update, and present the reference prices.
Recommendable products	It must be defined whether the first configurator covers only desktop computers or also standalone devices and peripherals.
Two alternatives	It must be defined which criteria differentiate options A and B, and what to do if there are not enough valid combinations.
External links	Links may become outdated; the system does not guarantee that third-party pages remain valid.
Academic time	The number of features and rules must be adjusted to the course's time frame and evaluation criteria.
Catalog access	It must be confirmed which queries will be available to unauthenticated visitors.

12. User stories

The user stories are organized according to the functional modules described in section 8. The priority indicates the importance of each story for the scope of the first version, and the Requirement column relates it to the initial functional requirements in section 9.

12.1. Authentication and users

ID	User story	Acceptance criteria	Priority	Requirement
US-01	**User registration** As a visitor, I want to create an account in HardwareMatch, so that I can access the hardware configuration features and save my recommendations.	● The system must allow registering a user with the required basic data. ● The system must validate that the required fields are complete and that the data meets the established rules. ● The system must prevent two accounts from being registered with the same email address. ● The password must be stored securely using a hashing algorithm. ● Upon successful	High	FR-01

ID	User story	Acceptance criteria	Priority	Requirement
		registration, the user must receive a confirmation that the account was created.		
US-02	**Login** As a registered user, I want to log in with my credentials, so that I can access the features authorized for my role.	• The system must request the credentials needed to log in. • The system must verify the credentials entered. • If the credentials are incorrect, it must report the error without revealing sensitive information. • If the credentials are correct, it must grant access corresponding to the user's role. • Protected features must not be available to users who have not logged in.	High	FR-01

ID	User story	Acceptance criteria	Priority	Requirement
US-03	**Profile viewing and updating** As a registered user, I want to view and update my basic data, so that I can keep my personal information up to date.	• The system must allow the authenticated user to view their profile data. • The system must allow updating the personal fields defined for the project. • The system must validate the data before saving changes. • Users must not be able to view or modify another user's profile through unauthorized operations.	Medium	FR-01, FR-14
US-20	**Role-based access control** As a system owner, I want features to be protected according to the user's role, so that unauthorized access to or modification of information is prevented.	• The system must differentiate the administrator and customer roles. • The administrator must be able to access the catalog and supplier	High	FR-02, FR-14

ID	User story	Acceptance criteria	Priority	Requirement
		management operations. ● The customer must be able to access the permitted browsing and recommendation features. ● The backend must validate permissions on protected operations. ● The system must reject requests that attempt to perform operations without authorization, even when made directly against the API.		

12.2. Catalog administration

ID	User story	Acceptance criteria	Priority	Requirement
US-04	**Product management** As an administrator, I want to create, view, update, and deactivate technology products, so that I can keep the HardwareMatch catalog up to date.	• The administrator must be able to register a product with its basic data. • The administrator must be able to view and modify the information of existing products. • The administrator must be able to deactivate a product so that it no longer appears as an available option in the catalog and in new recommendations. • The system must validate the required fields and the data associated with the product. • Users without administrative	High	FR-03

ID	User story	Acceptance criteria	Priority	Requirement
		permissions must not be able to perform product management operations.		
US-05	**Category management** As an administrator, I want to manage product categories, so that I can organize the catalog and make it easier to search for technology components.	• The administrator must be able to create and view categories. • The administrator must be able to update a category's information. • The system must validate that the category name is required and meets the established rules. • The system must allow relating products to their corresponding categories. • Categories must be available as catalog query criteria.	High	FR-03

ID	User story	Acceptance criteria	Priority	Requirement
US-06	**Brand management** As an administrator, I want to manage product brands, so that I can identify and classify the different options available in the catalog.	• The administrator must be able to create, view, and update brands. • The system must validate the required data for each brand. • The administrator must be able to associate a brand with the corresponding products. • Registered brands must be available for filtering catalog products.	Medium	FR-03
US-07	**Technical specification recording** As an administrator, I want to record and update the technical specifications of products, so that I provide useful information for comparison and compatibility validation.	• The system must allow recording the specifications defined for each product type. • The administrator must be able to view and update the registered	High	FR-03

ID	User story	Acceptance criteria	Priority	Requirement
		specifications. ● The system must validate the data required by the business rules. ● Specifications must be available for the details view and for the recommendation engine. ● If data needed to evaluate a compatibility is missing, the system must not assume that it exists.		
US-21	**Reference price management** As an administrator, I want to record and update the internal reference prices of products, so that the system can estimate the cost of hardware configurations.	● The system must allow recording a reference price for the products that require one. ● The administrator must be able to update the registered value. ● The system must	Medium	FR-03, FR-15

ID	User story	Acceptance criteria	Priority	Requirement
		validate that the price is a non-negative numeric value. ● The recommendation engine must use the registered reference prices to estimate costs. ● The system must distinguish these values from the commercial prices published by suppliers.		

12.3. Store and external link management

ID	User story	Acceptance criteria	Priority	Requirement
US-08	**Store and supplier management** As an administrator, I want to register and update stores or suppliers, so that I can associate them with the products available in the catalog.	• The administrator must be able to register a store with its name and the presentation data defined for the project. • The system must allow viewing and updating store information. • The system must allow registering a store's logo, according to the defined implementation. • Store information must be accessible from the related products.	Medium	FR-04
US-09	**External link association** As an administrator, I want to associate one or more external links with each product, so that I make it	• The administrator must be able to register external links for a product. • A product	Medium	FR-05

ID	User story	Acceptance criteria	Priority	Requirement
	easier for users to find additional information on the suppliers' pages.	must be able to have several links associated with different stores or suppliers. ● The system must validate that each link has a valid web address format. ● Users must be able to identify the supplier related to each link. ● Links must take the user to the registered external page. ● The system must state that availability and prices must be verified directly with the supplier.		

12.4. Catalog search and browsing

ID	User story	Acceptance criteria	Priority	Requirement
US-10	**Catalog**	● The system	High	FR-06

ID	User story	Acceptance criteria	Priority	Requirement
	browsing As a registered customer, I want to browse the technology product catalog, so that I know the available alternatives before selecting the components I need.	must display active, published products. ● Each product must show its basic information, category, brand, and reference price when registered. ● Users must be able to access a product's detail view. ● Deactivated products must not appear as available alternatives for new recommend ations. ● The listing must show understand able information that is consistent with the stored data.		
US-11	**Product search** As a registered customer, I want to search for products by text, so that I can quickly find the technology components I am	● The system must allow entering a search term. ● The search must query the text fields	Medium	FR-06

ID	User story	Acceptance criteria	Priority	Requirement
	interested in.	defined for the catalog, such as the product name. ● The system must display the products that match the criteria entered. ● If there are no matches, it must report that no results were found. ● The system must validate the search criteria in the backend.		
US-12	**Catalog filtering and sorting** As a registered customer, I want to filter and sort products by different criteria, so that I can compare alternatives that fit my preferences.	● The system must allow filtering products by the defined criteria, such as category, brand, and reference price range. ● Users must be able to combine several filters in a single query. ● The system must allow sorting	Medium	FR-07

ID	User story	Acceptance criteria	Priority	Requirement
		results by the implemented criteria, such as name, reference price, or registration date. ● The results must meet the selected filters and respect the chosen order. ● Users must be able to modify or remove the applied filters.		
US-13	**Catalog pagination** As a registered customer, I want to view the catalog results in pages, so that I can navigate through the products in an organized way.	● The system must split the results into pages according to the size defined for each query. ● Users must be able to navigate between the available pages. ● The system must indicate the current page or allow identifying the position	Medium	FR-08

ID	User story	Acceptance criteria	Priority	Requirement
		within the results. ● Pagination must work together with filters, search, and sorting. ● The backend must control pagination so that it does not depend exclusively on the interface.		

12.5. Configurator and recommendation engine

ID	User story	Acceptance criteria	Priority	Requirement
US-14	**Configuration requirements definition** As a registered customer, I want to indicate my budget and the intended use of the computer, so that I obtain hardware proposals tailored to my needs.	● The system must allow entering a maximum budget. ● The system must allow selecting an intended use from the options defined for the project. ● The system must validate that the budget is a positive numeric value. ● The system must validate that the data needed to generate the recommendation has been entered. ● The system must keep the request information when it is processed successfully.	High	FR-09
US-15	**Recommendation generation** As a registered	● The system must evaluate	High	FR-10, FR-11

ID	User story	Acceptance criteria	Priority	Requirement
	customer, I want the system to generate configuration proposals based on my budget and intended use, so that I can identify hardware alternatives suited to my needs.	candidate products using the selection rules defined by the team. ● The proposals must respect the established budget constraints. ● The system must consider the compatibility rules implemented for the components covered by the project. ● The system must use explicit and reproducible selection and scoring criteria. ● When enough data and valid combinations exist, the system must generate at least two alternatives. ● If the available information does not		

ID	User story	Acceptance criteria	Priority	Requirement
		allow valid proposals to be generated, the system must communicate the limitation to the user.		
US-16	**Hardware alternatives comparison** As a registered customer, I want to compare the generated configuration alternatives, so that I can identify their differences and select the one that best meets my needs.	● The system must present the alternatives generated for the same request. ● Each alternative must show the selected components and their main specifications. ● The system must show the estimated cost of each configuration. ● The system must make it possible to identify the relevant differences between the alternatives. ● The system	High	FR-11, FR-15

ID	User story	Acceptance criteria	Priority	Requirement
		must clarify that the prices used are internal reference values and do not necessarily match current commercial prices.		
US-17	**Configuration report viewing** As a registered customer, I want to view a detailed report of each recommended configuration, so that I understand the selected components and can review the information needed before making a decision.	• The report must present the components included in the configuration. • Each component must show its registered specifications and its reference price, when available. • The report must include the estimated cost of the configuration. • The system must present an explanation of the proposal based on the implemented selection	High	FR-12, FR-15

ID	User story	Acceptance criteria	Priority	Requirement
		criteria. ● The report must show the available external links to consult the products. ● The system must warn that availability, prices, and commercial conditions must be verified on the suppliers' pages.		

12.6. Reports and saved configurations

ID	User story	Acceptance criteria	Priority	Requirement
US-18	**Saving requests and configurations** As a registered customer, I want my recommendation requests and their results to be saved, so that I can later consult the configurations I have generated.	● The system must store the information of the processed request. ● The system must keep the relationship between the request and the generated alternatives. ● The stored information must make it possible to identify the components associated with each alternative. ● The system must maintain the relationship between each request and the user who made it. ● The data must persist after the user logs out or leaves the application.	High	FR-13

ID	User story	Acceptance criteria	Priority	Requirement
US-19	**Configuration history viewing** As a registered customer, I want to view my saved requests and configurations, so that I can review the results of previous recommendations without having to generate them again.	● The system must show the saved requests that belong to the authenticated user. ● Users must be able to select a request to view its results. ● The system must show the alternatives and components associated with the selected request. ● The system must not allow a user to view private configurations belonging to another user. ● If the user has no saved requests, the system must report that no records are available.	Medium	FR-13, FR-14